

MODULE 2.4 / 2.7 — NONCLINICAL AND CLINICAL OVERVIEWS

AMLOZA (Amlodipine Besylate Tablets, USP) 5 mg / 10 mg — ANDA

2.4 Nonclinical Overview

This is an Abbreviated New Drug Application (ANDA) submitted under Section 505(j) of the Federal Food, Drug, and Cosmetic Act. Amlodipine besylate is a well-established active ingredient, first approved in the United States in 1987 (Norvasc®, NDA 019787), with an extensive nonclinical and postmarketing safety database accumulated over more than three decades of clinical use.

Consistent with the ANDA pathway, no new nonclinical (pharmacology, pharmacokinetic, or toxicology) studies were conducted or are required to be submitted in support of this application. The nonclinical safety profile of amlodipine besylate is adequately characterized in the approved labeling of the Reference Listed Drug (RLD) and in the published pharmacological and toxicological literature, and this ANDA relies on FDA's prior finding of safety and effectiveness for the RLD.

Basis for Waiver of Nonclinical Studies

Regulatory basis	21 CFR 314.94(a)(8) — nonclinical studies are generally not required for an ANDA where the generic product does not raise new nonclinical safety questions relative to the RLD
Route/dosage form	Identical to RLD (oral tablet); no new route of administration introduced
Formulation risk assessment	Excipients used (Module 3.2.P.1) are standard, compendial, and previously used in FDA-approved oral solid dosage forms; no novel excipient or absorption-modifying component is introduced
Conclusion	No new nonclinical data are needed; the established nonclinical profile of amlodipine, as reflected in RLD labeling, is considered directly applicable to AMLOZA

2.7 Clinical Summary / Overview

As an ANDA, this application does not include independently generated clinical efficacy or safety studies. Clinical efficacy and safety for the amlodipine besylate active ingredient, at the proposed strengths and for the proposed indications, are established by the approved labeling of the RLD (Norvasc®). The clinical bridge between the RLD and AMLOZA is provided by the bioequivalence data presented in Module 5.

2.7.1 Clinical Pharmacology

Amlodipine's mechanism of action (inhibition of calcium ion influx across cardiac and vascular smooth muscle cell membranes, producing coronary and peripheral vasodilation) and its established pharmacokinetic profile (slow absorption, T_{max} 6–12 hours, extensive hepatic

metabolism, elimination half-life 30–50 hours supporting once-daily dosing) are as reported in the RLD's approved labeling, reproduced for reference in Module 1.11 (Full Prescribing Information). No independent clinical pharmacology data were generated for AMLOZA; the profile is relied upon from the RLD since AMLOZA contains the same active moiety at the same strengths.

2.7.2 Summary of Clinical Efficacy

Not independently established for this ANDA. Efficacy for the treatment of hypertension and chronic stable/vasospastic angina is relied upon from the RLD's approved labeling and the extensive published clinical literature on amlodipine besylate, consistent with the generic drug approval pathway.

2.7.3 Summary of Clinical Safety

Not independently established for this ANDA beyond the safety monitoring conducted within the bioequivalence study itself (Module 5). The established adverse reaction profile, warnings, and contraindications (see Module 1.11, Sections 4–8) are relied upon from the RLD's approved labeling.

2.7.4 Benefit-Risk Conclusion

The benefit-risk profile of AMLOZA would be expected to be comparable to that of the RLD (Norvasc®), based on demonstrated bioequivalence (Module 5), identical active ingredient and strengths, compendially controlled excipients, and reliance on FDA's prior finding of safety and effectiveness for the reference product.

Portfolio Note: These overviews demonstrate understanding of a core ANDA principle — generic applicants rely on FDA's prior finding of safety and efficacy for the RLD rather than generating independent nonclinical/clinical data, with bioequivalence serving as the scientific and regulatory bridge between the two products.